Using Youth Participatory Action Research to Address Inequities Surrounding Technology in the Math Classroom

EDPS 935

Maxi Project

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Abstract

In this proposal, I lay out a study based on the Youth Participatory Action Research (YPAR) epistemology, framework, and methodology to study the interaction of mathematics education and technology within a critical race perspective. Mathematics in the Western education context is often still considered completely immutable, objective, and True and this mindset can be incredibly dehumanizing to students as they are being required to learn something that is explicitly denying their knowledge and life experiences. The nature of YPAR demands that specific research questions, tools, methodologies, and implementations be designed within the context of the community selected. I will be working in two math classrooms at Westside Community Schools West Campus.

Introduction

How YPAR might be used to disrupt and transform traditional mathematics education

Specifically, within the context of technology access

Goal with this study is to propose a new way of looking at and teaching mathematics and incorporating technology

More inclusive to all students

Helps students challenge the dominant order to foment change beyond the classroom

Youth Participatory Action Research presents an entirely new(to the Western world) way of teaching, researching, and knowing that centers the students as the experts and leverages their individual, family, and cultural experiences to shape the learning taking place within a math classroom

Purpose Statement

The topic of this research proposal is how YPAR might be used to disrupt and transform traditional mathematics education, specifically within the context of technology access.

Key Terms

- YPAR
- De/colonialization
- Pláticas and counter-storytelling

Important Literature

No one wants to be the passive receiver of what someone else decides is best for them so why would we think that “educating” students that way would be effective?

YPAR presents a new way of teaching, learning, and knowing that actively and authentically involves students as stakeholders in their learning and works towards disrupting the dominant powers that currently work to oppress and marginalize communities that do not fit the “cultural ideal.”

YPAR and Counternarratives as Math Curriculum

YPAR throws out the traditional banking model of education and instead challenges educators to take a “step back and support the creative genius naturally ingrained in youth” (Gonzalez et al., 2017, p. 82).

This is especially important in the field of mathematics because it is often considered completely immutable, objective, and True and this mindset can be incredibly dehumanizing to students as they are being required to learn something that is explicitly denying their knowledge and life experiences.

PAR assumes that everyday people carry deep knowledge and analysis about the conditions of their lives and should lead in determining meaningful research questions, designs, methods, analyses and products” (Torre & Ayala, 2009, p. 388).

Mathematics and Technology

Technology has become ever-present in our world, and especially after the COVID-19 pandemic, technology has become a staple in K-12 classrooms.

Rubin (1999) wrote that "rather than looking at math education from the perspective of the computer, we must look at computers from the perspective of mathematics education" and continues saying, "the role of technology in the math classroom must be in service of goals we hold for students' mathematical knowledge and expertise" (p. 1).

What is clear from this small selection of research is that mathematics research is still quite centered around traditional Western epistemologies, and the research participants are passive objects to be studied, measured, and acted upon.

Mathematics and Student Motivation

Abu and Kribushi (2022) attempt to bridge the gap between the perceived lack of student motivation surrounding math utilizing technology after observing that, “the use of technology has not led to a revolution in teaching or learning” (p. 3).

This points to the effectiveness of the principles of YPAR, because it was not so much the technology that increased student motivation and academic achievement, but its appropriate implementation creating a more positive and collaborative learning environment.

The implication is that a collaborative learning environment focused on students’ funds of knowledge and using that knowledge for change is one of the most effective ways to cultivate student motivation and engagement in the classroom.

Conclusion

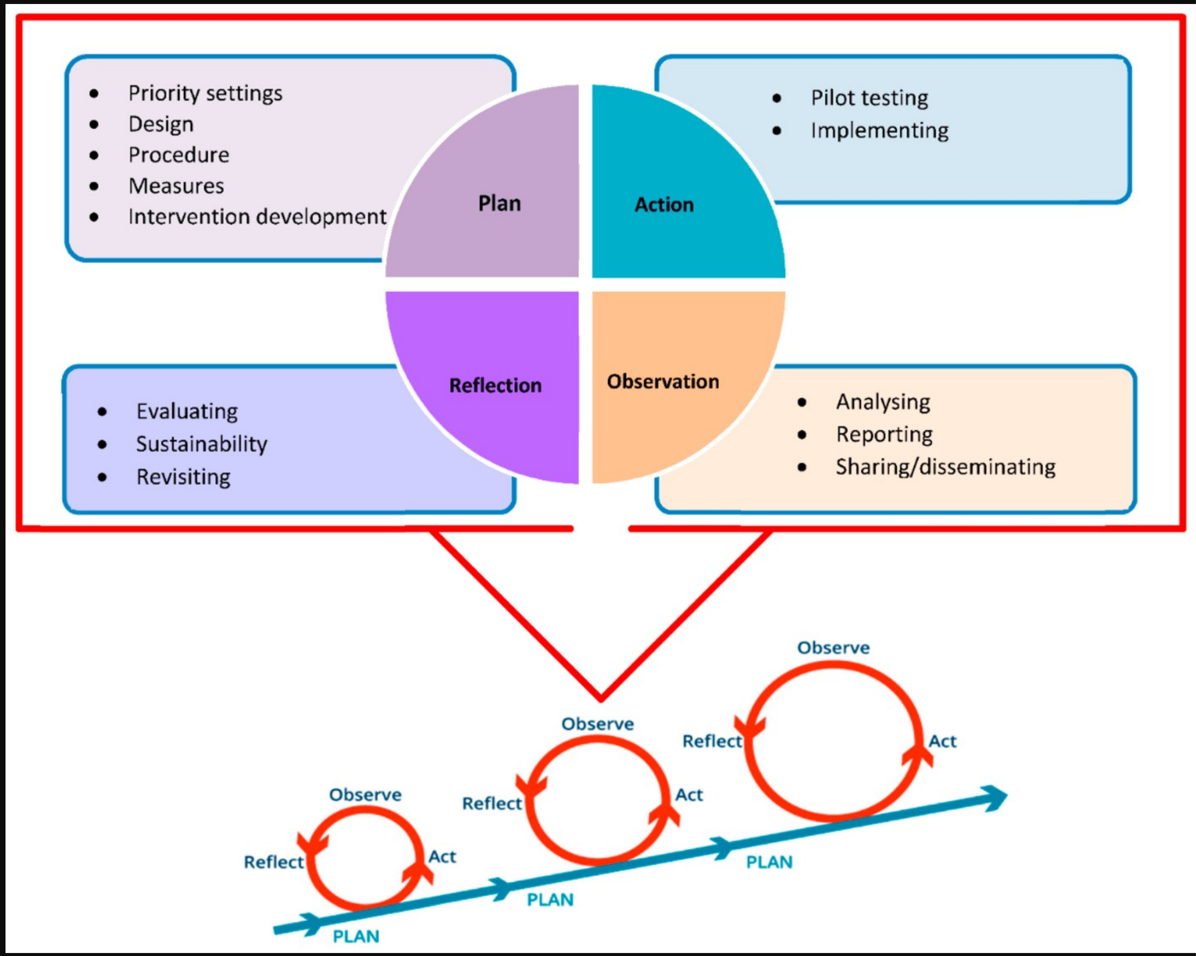
Overall, the state of mathematics educational research is still very quantitative in nature and “data” focused.

Youth Participatory Action Research has started to gain a foothold in mathematics education and shows promise as both an epistemology and ontology that can transform mathematics education.

Stavrou and Miller (2017) sum it up best by saying, “we believe that if researchers and educators acknowledge the root causes of inequality, then there would be a shift in the literature away from positing a lack of cultural relevance as the reason that Indigenous students are disengaged with mathematics” (p. 110).

Context & Framework

- YPAR is an approach that can work to re-humanize and de/colonialize both the mathematics classroom and also educational research.



Research Questions

The nature of YPAR necessitates that research questions be designed and written with the input of the stakeholders participating.

- Thus, while a tentative research question is proposed, it is understood that this may change over the course of the research from input from the participants.

What is the perceived impact of the intersection of mathematics education, technology, and racism within a middle or high school public school classroom?

- With YPAR being a collaborative process between students and researcher, additional sub-questions will be developed once researcher is in the classroom.

Research Design

YPAR is a cyclical process and is directed by the participants themselves. This means that much of the predetermined research approach will be focused on building trust and community within the chosen classroom. This will be accomplished by being present in the classroom as an active participant prior to the research taking place, providing education to the students and their parents about what the YPAR research will entail, and helping students develop the research questions that will guide all further research.

IRB and Ethical Considerations

- The IRB proposal submitted to UNL laid out any foreseen ethical challenges, how they will be addressed, and how any unforeseen ethical issues will be addressed as they arise. The proposal also described any potential benefits and risks to the participants and how they (and their guardians if necessary) will be made aware of these benefits and risks. Also provided in this proposal was the assent form and protocol documents to be used throughout the course of the research.
- IRB approval was received 11/15 (20241124016EP)

Methods of Investigation

Research Question	Methods	Analysis
What is the perceived impact of the intersection of mathematics education, technology, and racism within a middle or high school public school classroom?	Counter-storytelling Pláticas	Shared meaning making with subject checking of coding and summarizing of any counter-storytelling, Pláticas, or similar conversation-based data sources

Sample Selection

For the true pilot study:	Math Classroom in an urban school Sampling of convenience as access is limited due to me not currently teaching Westside Community Schools West Campus Students can opt in or out, and parental consent was granted by the district
For this class:	Sampling of convenience as access is limited due to me not currently teaching Two former students who I taught/mentored Siblings Parent is on my dissertation committee and a mentor friend

Data Collection

- Counter-storytelling
- Pláticas
- Class interviews
- Conversations are recorded via Zoom and transcribed by Zoom as a starting point for the final transcript

Data Analysis

Thematic Analysis Coding

Braun & Clarke

Validated transcript with audio file while taking notes and initial impressions

Pulled out several reoccurring themes:

COVID	Paper/writing is better than typing	Can’t connect to content	HAL also an access issue

“We imagine qualitative analysis as an adventure, and one that is typically messy and organic, complex and contested—’more messy, more convoluted and more surprising than we thought it would be’”

“I feel a little bit old saying this, but I feel like sometimes computers, especially for working with the new system or something, can be confusing”

“Jim”

“Sometimes, not necessarily. It’s not necessarily like sometimes the old ways. Were working for a reason. Like it’s okay. It’s okay to go back to kind of paper like you’re not… You’re not…The trees will be fine. There’s worse people in the world. So, if you need to do your work on paper, do you.”

“AJ”

Findings

- Students felt technology was often introduced because it was “cool” not because it would actually enhance learning
- Both students talked about missing typing and computer literacy classes in elementary school when pulled out for HAL services
- Writing out your thoughts versus typing them helps content “stick” better
- Overall, felt technology implementation needed to be more intentional

Anticipated Products of Pilot

Over the course of this study, I hope to develop a fuller picture of the intersection of mathematics education, technology, and racism in the upper-level, public school classroom.

The results of this specific study may have limited generalizability as each classroom, school, district, and community faces varying challenges and obstacles differently, even if they are facing the "same" challenges and obstacles as another community. However, the concepts of implementing YPAR in this study are generalizable across communities as the goal of YPAR is to provide a framework that is stakeholder driven and focused on actionable solutions for the specific community in question.

Rigor Criteria	Purpose	Original Strategies	Strategies applied in our study to achieve rigor
Credibility	To establish confidence that the results (from the perspective of the participants) are true, credible, and believable.	• Prolonged and varied engagement with each setting	• Researchers will spend an average of 6–8 weeks per site to engage with participants.
		• Establishing investigators' authority	• We will ensure the investigators had the required knowledge and research skills to perform their roles.
		• Collection of referential adequacy materials	• We will ask interviewers to send all the field notes to the research office for analysis and storage.
		• Peer debriefing	• We will have regular debriefing sessions with key members of the project
Dependability	To ensure the findings of this qualitative inquiry are repeatable if the inquiry occurred within the same cohort of participants, coders, and context.	• Rich description of the study methods	• We will prepare detailed drafts of the study protocol throughout the study.
		• Establishing an audit trail	• We will develop a detailed track record of the data collection process.
		• Stepwise replication of the data	• We plan to measure coding accuracy and inter-coders' reliability of the research team.

Credibility and Transferability

Rigor Criteria	Purpose	Original Strategies	Strategies applied in our study to achieve rigor
Confirmability	To extend the confidence that the results would be confirmed or corroborated by other researchers.	• Reflexivity	• We will implement reflexive journals and weekly investigators meetings.
		• Triangulation	• We will apply several triangulation techniques (methodological, data source, investigators and theoretical).
Transferability	To extend the degree to which the results can be generalized or transferred to other contexts or settings.	• Purposeful sampling to form a nominated sample	• We will use a combination of three purposive sampling techniques.

Credibility and Transferability